

Image Classification SOTA

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1. Introduction to Deep Convolutional Neural Networks

- Our network has 60 million parameters and 650,000 neurons with five convolutional layers and three fully-connected layers
- We achieved top-1 and top-5 error rates of 37.5% and 17.0% on ImageNet, a significant improvement over previous methods
- Used ReLU nonlinearity for faster training and dropout regularization to reduce overfitting
- Implemented overlapping pooling and local response normalization to improve performance
- Trained on two GPUs with 3GB memory each, achieving 15.3% top-5 error rate in ILSVRC-2012 competition

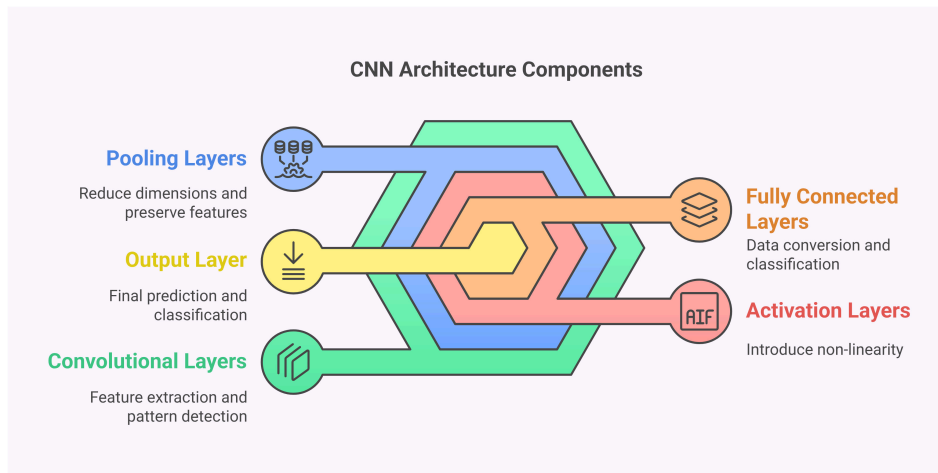
1.1. Background and Motivation

- Our network has 60 million parameters and 650,000 neurons with 5 convolutional layers and 3 fully-connected layers
- We achieved top-1 and top-5 error rates of 37.5% and 17.0% on ImageNet, outperforming previous state-of-the-art methods
- Used ReLU nonlinearity for faster training and dropout regularization to reduce overfitting
- Implemented overlapping pooling and local response normalization to improve generalization
- Trained on two GPUs with 3GB memory each, taking 5-6 days to train on 1.2 million images

1.2. Key Contributions of the Paper

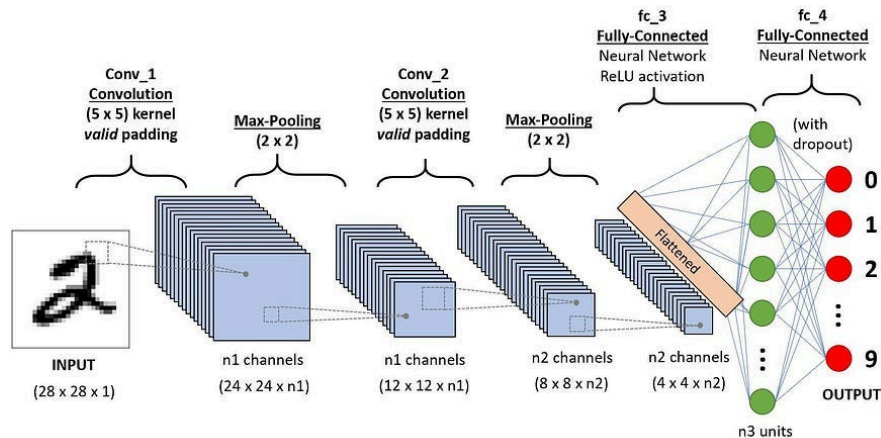
- Our paper trained one of the largest CNNs to date on ImageNet, achieving record-breaking error rates of 37.5% (top-1) and 17.0% (top-5) on ILSVRC-2010.
- We introduced a highly-optimized GPU implementation of 2D convolution, making large-scale CNN training feasible for the first time.
- Key innovations include ReLU nonlinearity (training 6x faster than tanh), overlapping pooling (reducing error by 0.4%), and local response normalization (reducing error by 1.4%).
- To combat overfitting, we used dropout (doubling training time but preventing complex co-adaptations) and data augmentation (translating images and altering RGB intensities via PCA).

2. Architecture and Design of the CNN Model



- Our CNN has 60 million parameters and 650,000 neurons with 5 convolutional layers and 3 fully-connected layers
- We used ReLU nonlinearity for faster training compared to tanh or sigmoid
- Training on multiple GPUs allowed us to scale the network size beyond single-GPU limitations
- Local response normalization reduced top-1 and top-5 error rates by 1.4% and 1.2%
- Overlapping pooling ($s=2$, $z=3$) reduced error rates by 0.4% and 0.3% respectively

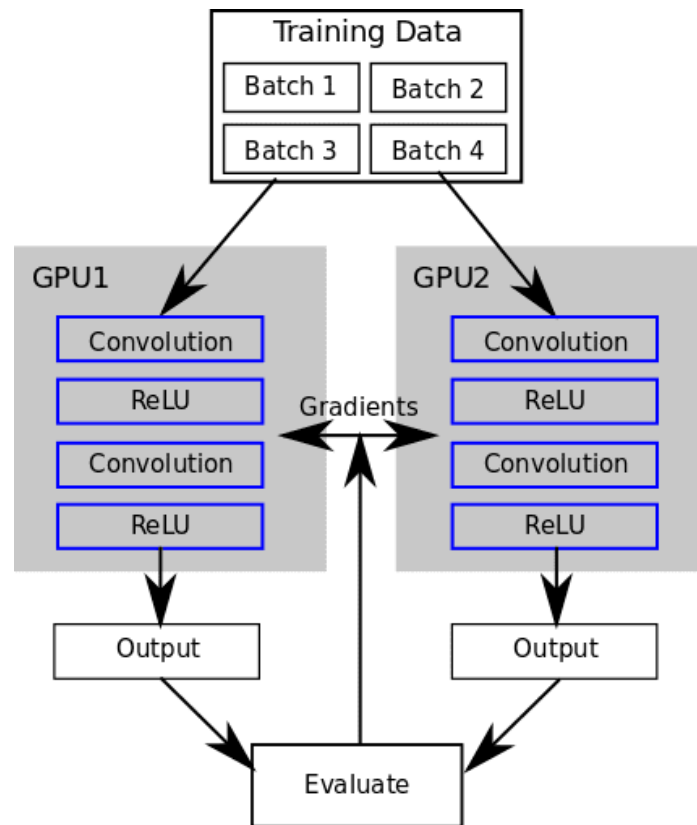
2.1. Core Components of the Network



- Our CNN has 8 learned layers: 5 convolutional and 3 fully-connected, with 60 million parameters.
- The first convolutional layer uses 96 $11 \times 11 \times 3$ kernels with stride 4 to process $224 \times 224 \times 3$ images.
- ReLU nonlinearity accelerates training by several times compared to tanh units, as shown in Figure 1.
- Local response normalization reduces top-1 and top-5 error rates by 1.4% and 1.2% respectively.
- Overlapping pooling ($s=2$, $z=3$) reduces error rates by 0.4% and 0.3% compared to non-overlapping pooling.

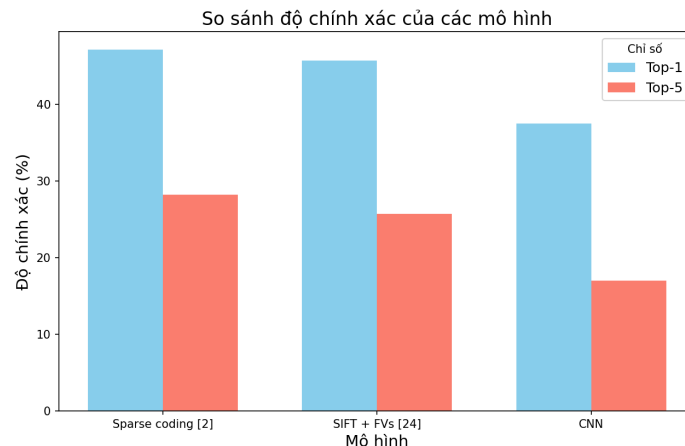
2.2. Innovative Features for Performance Improvement

- ReLU nonlinearity accelerates training by several times compared to tanh, as shown in Figure 1 with a four-layer CNN on CIFAR-10.
- Using ReLUs allows faster learning, which is critical for large models trained on big datasets like ImageNet.
- Our network uses two GPUs to handle larger models, with communication only in specific layers to reduce overhead.
- Local response normalization improves generalization with constants $k=2$, $n=5$, $\alpha=10^{-4}$, $\beta=0.75$, reducing error rates by 1.4% and 1.2%.
- Overlapping pooling ($s=2$, $z=3$) reduces top-1 and top-5 error rates by 0.4% and 0.3% compared to non-overlapping schemes.



3. Training and Optimization Techniques

- We trained a large, deep CNN with 60 million parameters to classify 1.2 million ImageNet images into 1000 classes.
- Our network achieved top-1 and top-5 error rates of 37.5% and 17.0%, significantly better than previous state-of-the-art methods.
- To accelerate training, we used ReLU nonlinearity, which trains several times faster than traditional saturating nonlinearities like tanh.
- We implemented a highly optimized GPU convolution that enabled training of large networks on limited hardware.
- To reduce overfitting, we employed dropout and data augmentation techniques that improved generalization.

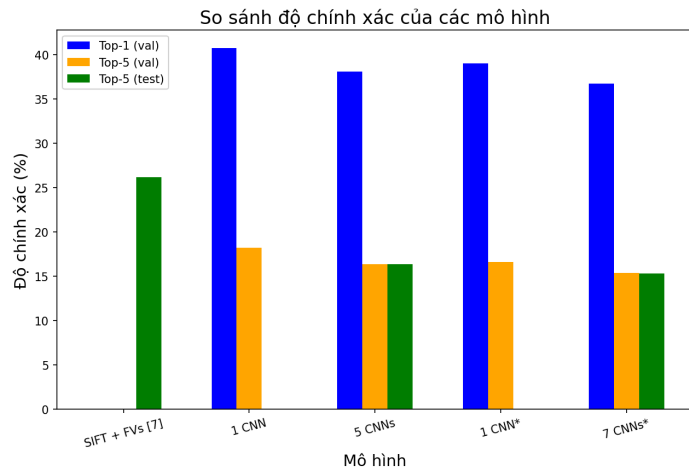


3.1. Data Augmentation Strategies

- Data augmentation is crucial for reducing overfitting in large CNNs with 1.2 million training examples.
- We use two forms of data augmentation: random image translations & horizontal reflections (2048x larger dataset) and PCA-based color jittering.
- For translations, we extract random 224×224 patches from 256×256 images and their horizontal reflections, increasing training set size by 2048x.
- For color augmentation, we perform PCA on RGB pixel values and add multiples of principal components with magnitudes proportional to eigenvalues times Gaussian noise ($\sigma=0.1$).
- This scheme captures illumination invariance and reduces top-1 error rate by over 1%.

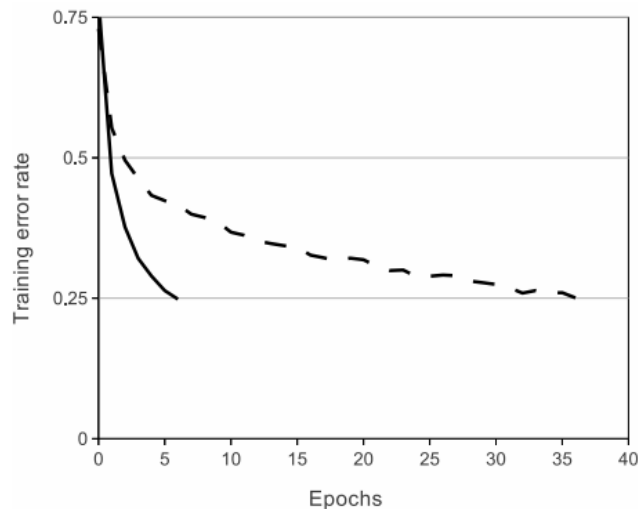
4.1. Performance on ImageNet Datasets

- Our network achieved top-1 and top-5 error rates of 37.5% and 17.0% on ILSVRC-2010 test set
- This outperformed previous state-of-the-art methods like sparse coding (47.1% top-1) and SIFT + FVs (45.7% top-1)
- With ensemble methods, we reached 15.3% top-5 error rate - a winning result in ILSVRC-2012
- The network has 60 million parameters with 5 convolutional and 3 fully-connected layers
- We used ReLUs, dropout, data augmentation, and GPU parallelization to combat overfitting



4.2. Comparative Analysis with Other Models

- Our network achieved top-1 and top-5 error rates of 37.5% and 17.0% on ILSVRC-2010 - a significant improvement over previous state-of-the-art methods.
- Compared to the second-best entry in ILSVRC-2012, our model achieved a top-5 error rate of 15.3% vs. 26.2% - demonstrating superior performance.
- Using dropout and data augmentation techniques helped reduce overfitting, with dropout roughly doubling training iterations needed for convergence.
- Local response normalization reduced top-1 and top-5 error rates by 1.4% and 1.2% respectively, while overlapping pooling improved results by 0.4% and 0.3%.



Thanks for listening!

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